

A Delayless Subband Active Noise Control System for Wideband Noise Control

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Abstract—In this paper, we present an efficient noise control algorithm based on the delayless subband adaptive filter. The algorithm reduces the computational complexity of the delayless subband filter by decomposing the secondary path response into a set of subband functions. In this new algorithm, the filtered reference signal is generated for each subband by using a short impulse response filter that models the secondary path transfer function in a subband-decomposed form. Computational efficiency of the presented method originates from the fact that the filtering process occurs only in one subband for each reference input sample. Furthermore, computational efficiency is enhanced when the presented algorithm is combined with an online identification of the secondary path transfer function. We also propose a frequency-domain implementation of the presented algorithm. In this implementation, it is shown that the computational complexity is further reduced by employing the block-processing approach.

Index Terms—Active noise control (ANC), delayless subband adaptive filter, frequency-domain adaptive filter, online system identification, wideband noise.

I. INTRODUCTION

WIDEBAND active noise control (ANC) requires hundreds of taps to realize significant cancellation of a disturbance [1]. In addition to high computational complexity, adaptive filters with many taps suffer from slow convergence, especially, when the eigenvalues of the underlying correlation matrix of the input signal are widely spread [2].

Recently, new techniques based on the subband adaptive filter have been developed to reduce computational complexity and to improve the convergence speed [3]–[5]. One major disadvantage of such subband adaptive filters is a delay introduced into the signal path by virtue of the bandpass filtering necessary to derive the subband signals. In ANC problems, the delay is a factor that seriously limits the bandwidth over which good cancellation can be achieved [6].

The delayless subband adaptive filter [7] suggested by Morgan and Thi computes the adaptive weights in subbands, but collectively transforms them into an equivalent set of wideband filter coefficients. In such a way, signal path delay

can be avoided while retaining the advantages of subband processing. An additional benefit of this approach accrues through a significant reduction of aliasing effects. However, since the filtered-X LMS (FXLMS) algorithm [2], [8] compensates the phase distortion of the secondary path by filtering the reference signals with a wideband filter modeling the secondary path, computational demands of the wideband filtering can be significantly increased in some cases, in particular, when the secondary path is modeled with a long impulse response filter.

In this paper, we present an efficient noise control algorithm based on the delayless subband adaptive filter architecture. In the new algorithm, the computational complexity is greatly reduced by decomposing the secondary path transfer function into a set of subband functions. In contrast to the method proposed by Morgan and Thi [7], the filtered reference input is generated for each subband by using short impulse response filters that model the secondary path transfer function in a decomposed form. The filtering process in the presented algorithm occurs in just one subband per reference input sample. Furthermore, the computational merit of this approach is enhanced when it is combined with an online system identification of the secondary path [9], [10].

We also propose a frequency-domain implementation scheme for the algorithm. In this implementation, the convolution operation for the reference signal filtering is replaced with a multiplication of frequency coefficients. It is shown that the frequency-domain implementation is capable of reducing the computational load further by employing the block-processing approach. Computer simulations were performed to evaluate the efficiency of the ANC algorithm and the online system identification scheme presented here.

This paper is organized as follows. Section II introduces a new delayless subband ANC system using a subband-decomposed secondary path transfer function. An efficient frequency-domain implementation of the proposed system is presented in Section III and computational requirements of the new ANC system are discussed in Section IV. Simulation results of the proposed ANC system are presented in Section V.

II. DELAYLESS SUBBAND ANC SYSTEM WITH SUBBAND-DECOMPOSED SECONDARY PATH TRANSFER FUNCTION

In order to assure the stability and convergence of the adaptive ANC filter, the reference signal is filtered using an estimate of the impulse response of the secondary path [7]. Thus, as the number of coefficients used to model the secondary path response grows, computational demands for the filtering process become significant. This difficulty can be overcome by taking

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advantage of the structure of the subband filter. Instead of generating the filtered reference signal through filtering over the entire band of interest, the filtered reference signal can be obtained for each subband. To this end, the secondary path transfer function can be decomposed into a set of subband functions. Fig. 1 shows the structure of the delayless subband ANC system in which the secondary path transfer function is decomposed, and then, applied to each subband. The block $P(z)$ in Fig. 1 represents the transfer function from the noise source to the error sensor S , and $C(z)$ denotes the transfer function from the adaptive filter output to the error sensor S .

Except for the process of reference input filtering, the implementation of delayless subband system is in parallel to the method described in [7]. The polyphase FFT technique [4] can be used to generate subband input signals. This technique realizes M contiguous single-sideband bandpass filters whose outputs are downsampled by a factor $D = M/2$ to produce M complex subband signals. In this paper, the $2\times$ over-sampled polyphase FFT subband filtering technique is utilized. In each subband, the signals are decimated by a factor of D and the filter weights are adjusted using the complex normalized LMS (NLMS) algorithm [11]. The filter weights in each subband are then transformed into frequency-domain coefficients using the FFT, after appropriately stacked, and inversely transformed to obtain the wideband filter coefficients. For real signals, the wideband filter coefficients are real and only half of the subbands, corresponding to the positive frequency components of the wideband filter response, need to be processed. The other half of the response is formed in complex conjugate symmetry.

Assuming that the transfer function of the secondary path, denoted by $C(z)$ in Fig. 2, is modeled as an L th order FIR filter, and denoting $c_{m,l}$, $0 \leq l \leq L/D - 1$ as the FIR coefficients of $C(z)$ corresponding to the subband m , the filtered reference signal $r_m(n)$ of subband m is expressed as

$$r_m(n) = \sum_{l=0}^{L/D-1} c_{m,l} x_m(n-l),$$

$$m = 0, 1, \dots, D, \quad n = 0, D, 2D, \dots \quad (1)$$

where $x_m(n)$ is the reference input.

For an N th-order ANC filter with a decimation factor D , the FXLMS update for the m th subband is expressed as

$$\mathbf{w}_m(n+D) = \mathbf{w}_m(n) - \frac{\mu}{\|\mathbf{r}_m(n)\|^2} \mathbf{e}_m^*(n) \mathbf{r}_m(n),$$

$$m = 0, 1, \dots, k, \dots, D \quad (2)$$

where $\mathbf{w}_m(n) = [w_{m,0}(n), w_{m,1}(n), \dots, w_{m,N/D-1}(n)]^T$ is a (N/D) -length subband weight vector, $\mathbf{r}_m(n) = [r_m(n), r_m(n-1), \dots, r_m(n-N/D+1)]^T$ is the filtered reference input vector of the m th subband, and $e_m(n)$ is the error signal. Superscript $*$ in (2) denotes complex conjugate and $\|\cdot\|$ indicates the Euclidean norm operator of the vector.

The delayless subband ANC system in [7] assumes off-line modeling of the secondary path. In some ANC problems, however, due to continuous changes of the secondary path, it is necessary to identify the impulse response of the secondary path

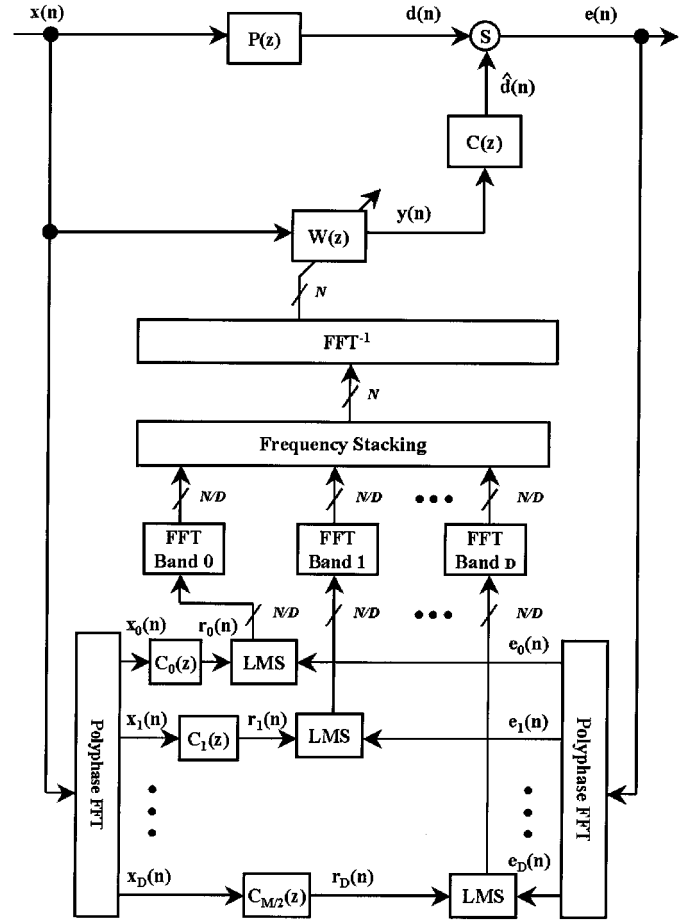


Fig. 1. Delayless subband ANC system with a subband-decomposed secondary path transfer function.

during the real-time operation in order to assure the stability and convergence of the adaptive filter. Previously, Yun *et al.* suggested that the delayless subband ANC system combine with wideband online system identification [12].

In [9], the two filter online modeling algorithm is applied to each subband to estimate the secondary path response in a decomposed form. Fig. 2 shows a schematic diagram of the delayless ANC system comprising the online system identification filter implemented in the m th subband.

Defining $(L \times 1)$ coefficient vectors of the primary and secondary path system identification filters for the m th subband as $\hat{\mathbf{p}}_m(n)$ and $\hat{\mathbf{c}}_m(n)$, respectively, they are adjusted to minimize the power of the estimation error, represented by $\tilde{e}_m(n)$, given by

$$\tilde{e}_m(n) = e_m(n) - [\hat{\mathbf{p}}_m^H(n) \mathbf{x}_m(n) + \hat{\mathbf{c}}_m^H(n) \mathbf{y}_m(n)] \quad (3)$$

where $\mathbf{x}_m(n)$ and $\mathbf{y}_m(n)$ are the $(L \times 1)$ input and output vectors of the m th subband filter. The weight vectors can then be updated using the NLMS algorithm as

$$\hat{\mathbf{p}}_m(n+D) = \hat{\mathbf{p}}_m(n) + \frac{\kappa}{\|\mathbf{x}_m(n)\|^2 + \|\mathbf{y}_m(n)\|^2} \tilde{e}_m^*(n) \mathbf{x}_m(n) \quad (4)$$

$$\hat{\mathbf{c}}_m(n+D) = \hat{\mathbf{c}}_m(n) + \frac{\kappa}{\|\mathbf{x}_m(n)\|^2 + \|\mathbf{y}_m(n)\|^2} \tilde{e}_m^*(n) \mathbf{y}_m(n). \quad (5)$$

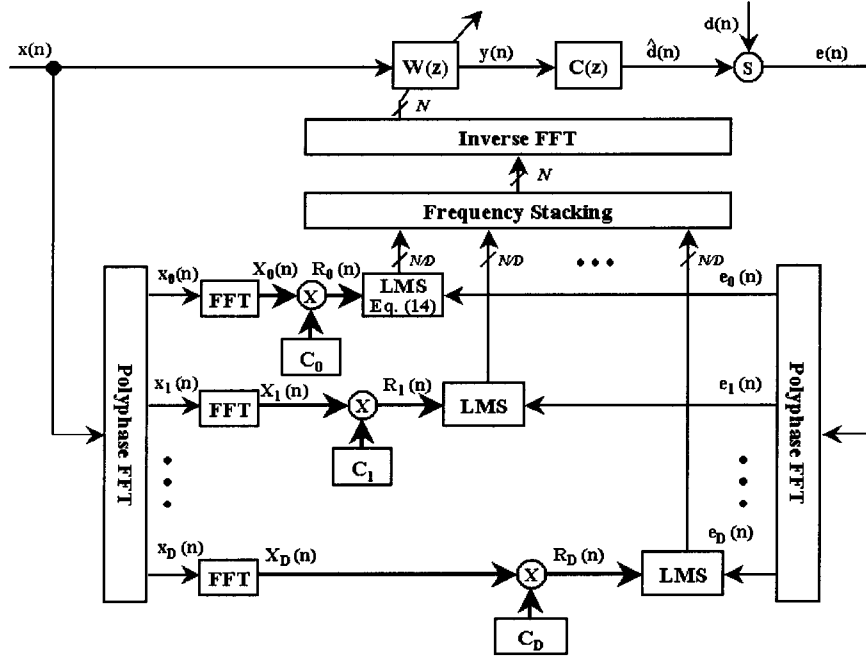


Fig. 3. Delayless subband ANC system with the frequency-domain update scheme.

Now, the complex NLMS algorithm for updating $\mathbf{W}_m(n)$ can be obtained as

$$\begin{aligned}\mathbf{W}_m(n+D) &= \mathbf{W}_m(n) - \frac{\mu'}{\|\mathbf{R}_m(n)\|^2} e_m^*(n) \mathbf{R}_m(n) \\ &= \mathbf{W}_m(n) - \frac{\mu'}{\|\mathbf{R}_m(n)\|^2} e_m^*(n) \mathbf{\Lambda}_m \mathbf{X}_m(n), \\ m &= 0, 1, \dots, D \quad (13)\end{aligned}$$

where the convergence parameter is normalized by $\|\mathbf{R}_m(n)\|$ instead of $\|\mathbf{r}_m(n)\|$ based on Parseval's theorem, so that $\mu' = \mu/(N/D)$.

In (13), the secondary path response was assumed to be known *a priori*. In that case, the matrix $\mathbf{\Lambda}_m$ can be computed before the ANC system is on. Thus, when the ANC system is turned on, the input data vector $\mathbf{x}_m(n)$ is transformed and multiplied by $\mathbf{\Lambda}_m$ to obtain the transformed filtered reference input vector $\mathbf{R}_m(n)$.

Fig. 3 shows a block diagram of the delayless subband ANC system, in which the coefficients of the adaptive filters are updated using (13). In this new structure, the filtered reference signal is obtained by multiplication in the frequency-domain, rather than convolution in time-domain, which results in a significant reduction of the computational complexity required to generate the filtered reference signal. Also, it should be noted that the structure in Fig. 3 can be combined with the online identification shown in Fig. 2 without difficulty.

In the frequency-domain LMS algorithm, constraints are often imposed in order to implement strictly linear convolution between the filter input and impulse response. However, these constraints are not always desirable because we can achieve a simpler adaptive filter that can implement either linear or circular convolution by removing the constraints. Note that, by allowing the circular convolution to the filtered reference input, it is still possible for the adaptive filter to minimize the

mean-square error [17]. Furthermore, the fast convolution used in the structure Fig. 3 is only to update the adaptive weights, and the filter output is generated using the time-domain linear convolution.

The frequency-domain adaptive algorithms often have a recursion that is based the block update [5]. In the sliding DFT [18] and the frequency sampling [19] algorithms, the adaptive recursion operates at the incoming data rate. Although they exploit the orthogonality properties of the DFT, it is impractical to implement due to their computational disadvantages. By using the block update scheme, the NLMS update in (13) becomes

$$\mathbf{W}_m(n+L) = \mathbf{W}_m(n) - \frac{\mu'}{\|\mathbf{R}_m(n)\|^2} e_m^*(n) \mathbf{R}_m(n), \quad m = 0, 1, \dots, D. \quad (14)$$

This expression represents a single update of the weights based on the $N(=(2N/M) \times (M/2))$ data samples accumulated instead of $M/2$ samples.

IV. COMPUTATIONAL COMPLEXITY

The delayless subband ANC algorithm can be subdivided into five parts:

- 1) subband filtering;
- 2) adaptive weight update;
- 3) subband/wideband weight transformation;
- 4) signal path convolution with wideband filter W ;
- 5) filtered reference signal generation.

In this section, the computational complexity (C) of the proposed algorithms is measured with the number of real multiplications per input sample.

Let us assume that the adaptive filter has N taps. Since each $2 \times$ oversampled polyphase FFT subband filtering requires one K -tap prototype convolution and one M -point FFT per $M/2$

TABLE I

COMPUTATIONAL COMPLEXITY BASED ON THE NUMBER OF REAL MULTIPLICATIONS PER INPUT SAMPLE: (a) THE FULLBAND FXLMS STRUCTURE, (b) THE DELAYLESS SUBBAND ANC STRUCTURE SHOWN IN FIG. 1, (c) THE FREQUENCY-DOMAIN STRUCTURE SHOWN IN FIG. 3

Operation	(A)	(B)	(C)
C_1 : subband filtering	none	$\frac{4K}{M} + 4\log_2 M$	$\frac{4K}{M} + 4\log_2 M$
C_2 : weight updates	N	$\frac{8N}{M} + \frac{16N}{M^2}$	$4 + \frac{8}{M}$
C_3 : weight transform into wideband filter	none	$2\log_2 \frac{2N}{M} + \frac{4}{M}\log_2 \frac{2N}{M} + \log_2 N$	$\log_2 N$
C_4 : wideband filtering	N	N	N
C_5 : generating filtered-X signal	L	$\frac{8L}{M} + \frac{16L}{M^2}$	$4 + \frac{8}{M} + 2\log_2 \frac{2N}{M} + \frac{4}{M}\log_2 \frac{2N}{M}$
$C_{total} = \sum_{i=1}^5 C_i$	$2N + L$	$N + \frac{4(K+2N+2L)}{M} + \frac{16(N+L)}{M^2} + 2\log_2 M + 3\log_2 N + \frac{4}{M}\log_2 \frac{2N}{M}$	$8 + N + \frac{4(4+K)}{M} + 4\log_2 M + \log_2 N + (2 + \frac{4}{M})\log_2 \frac{2N}{M}$
C_6 : estimating 2nd path (optional)	$4L$	$\frac{40L}{M} + \frac{80L}{M^2}$	$\frac{40N}{M} + \frac{80N}{M^2}$
$C_{total} = \sum_{i=1}^6 C_i$	$2N + 5L$	$N + \frac{4(K+2N+12L)}{M} + \frac{16(N+6L)}{M^2} + 2\log_2 M + 3\log_2 N + \frac{4}{M}\log_2 \frac{2N}{M}$	$8 + N + \frac{4(4+K+10N)}{M} + \frac{80N}{M^2} + 4\log_2 M + \log_2 N + (2 + \frac{4}{M})\log_2 \frac{2N}{M}$

input samples, the subband filtering for the reference input and error signals requires

$$C_1 = \frac{2 \times (K + M \log_2 M)}{M/2} = \frac{4K}{M} + 4\log_2 M \quad (15)$$

real multiplications per input sample. For the real signals, only $(M/2) + 1$ complex subbands need to be processed. As a results of downsampling by a factor of $D = M/2$, it is necessary to update $2N/M$ complex weights per $M/2$ input samples. Thus, the LMS update for the structure in Fig. 2 requires

$$C_2 = \frac{4 \times (2N/M) \times (M/2 + 1)}{M/2} = \frac{8N}{M} + \frac{16N}{M^2} \quad (16)$$

real multiplications per input sample.

By using the structure shown in Fig. 3, the multiplication figure becomes much smaller than the structure in Fig. 2. Since the weights are updated for each batch of $(2N/M)$ subband samples, the LMS update in this case requires only

$$C'_2 = \frac{4 \times (2N/M) \times (M/2 + 1)}{(M/2) \times (2N/M)} = 4 + \frac{8}{M} \quad (17)$$

real multiplications per input sample.

In order to transform the subband weights into fullband weights, $(2N/M)$ -point complex FFT is required for each of $(M/2) + 1$ lower subbands. Later, the weights are stacked and transformed using an N -point complex FFT. Thus, in the structure in Fig. 1, the weight transformation process requires

$$\begin{aligned} C_3 &= \frac{[(\frac{M}{2} + 1) \times (\frac{4N}{M} \log_2 \frac{2N}{M}) + N \log_2 N]}{N} \\ &= 2\log_2 \frac{2N}{M} + \frac{4}{M} \log_2 \frac{2N}{M} + \log_2 N \end{aligned} \quad (18)$$

real multiplications per sample. On the other hand, the frequency-domain approach shown in Fig. 3 requires only an N -point complex FFT since the subband weights are already in

the frequency-domain. Thus, the number of real multiplications per sample becomes

$$C'_3 = \log_2 N. \quad (19)$$

Assuming that the secondary path response is modeled with an L th order FIR filter, the delayless subband ANC system in [7] requires L real multiplications to generate a reference sample. In the structure shown in Fig. 1, each subband uses $2L/M$ coefficients for the secondary path modeling. Also, $(M/2 + 1)$ filterings are required for each block of $M/2$ input samples to generate the filtered reference samples of lower $(M/2 + 1)$ subbands. Thus, the required number of real multiplications is

$$C_5 = \frac{4 \times (2L/M) \times (M/2 + 1)}{M/2} = \frac{8L}{M} + \frac{16L}{M^2}. \quad (20)$$

For the structure in Fig. 3, the reference input of each subband is transformed first via the $(2N/M)$ -point FFT and multiplied by $(2N/M)$ frequency coefficients representing the subband region of the secondary path transfer function. Thus, generating the filtered reference input in this case requires

$$\begin{aligned} C'_5 &= \frac{(\frac{M}{2} + 1) \times (\frac{4N}{M} \log_2 \frac{2N}{M} + 4 \times \frac{2N}{M})}{(\frac{M}{2} \times \frac{2N}{M})} \\ &= 4 + \frac{8}{M} + 2\log_2 \frac{2N}{M} + \frac{4}{M} \log_2 \frac{2N}{M} \end{aligned} \quad (21)$$

real multiplications per sample.

When the two-filter online modeling algorithm is used, $8L/M$ real multiplications for the filter output $y_m(n)$ and $32L/M$ real multiplications for the two adaptive filters $\hat{\mathbf{p}}_m(n)$ and $\hat{\mathbf{c}}_m(n)$ are required. Thus, the number of real multiplications for the online modeling is

$$C_6 = \frac{(40L/M) \times (M/2 + 1)}{M/2} = \frac{40L}{M} + \frac{80L}{M^2}. \quad (22)$$

Computational complexities of the three algorithms are summarized in Table I. Note that C_4 could be further reduced by

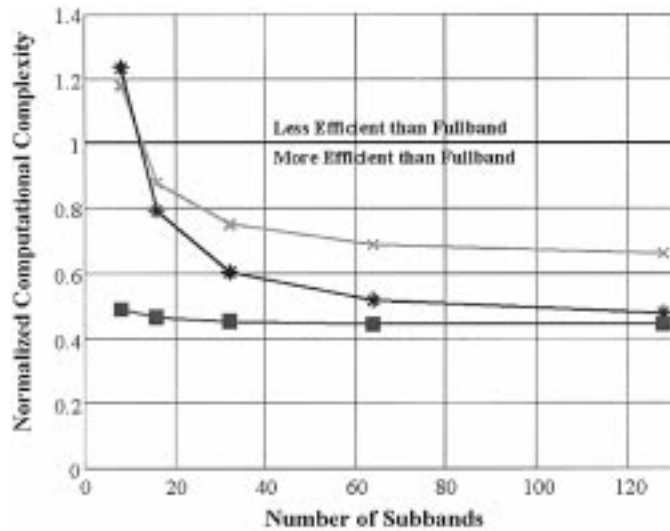


Fig. 4. Normalized computational complexity based on the number of real multiplications. $N = 512$, $L = 256$, and $K = 128$ were used for all cases. (x: Morgan and Thi [7], *: Fig. 1, ■: Fig. 3.)

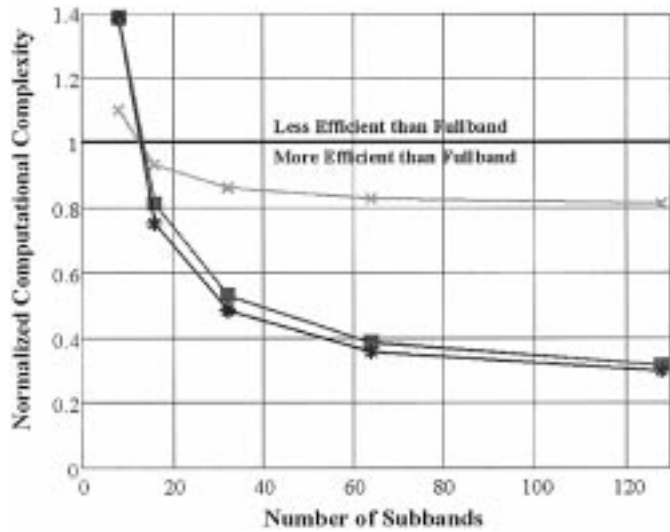


Fig. 5. Normalized computational complexity based on the number of real multiplications with the online identification of the secondary path. $N = 512$, $L = 256$, $K = 128$ were used for all cases. (x: Morgan and Thi [7], *: Fig. 1, ■: Fig. 3.)

using the technique in [7] for implementing the wideband signal path convolution.

Normalized computational complexities based on the number of real multiplications are presented in Figs. 4 and 5. Normalized computational complexity is defined as a ratio of the total number of real multiplications of each ANC system to that of the fullband structured ANC system (Table I(A)) per reference input sample. For these examples, we assumed $N = 512$, $L = 256$, and $K = 128$. Figure 4 shows the computational complexity with a pre-estimated secondary path and Fig. 5 shows the computational complexity including the online estimation of the secondary path. In the figures, x is for the system proposed in [7], * indicates for the system in Fig. 1, and ■ is for the system in Fig. 3. Both Figs. 4 and 5 clearly show that the computational benefits of the proposed ANC system become more significant as the number of subbands increases ($N > 16$). For

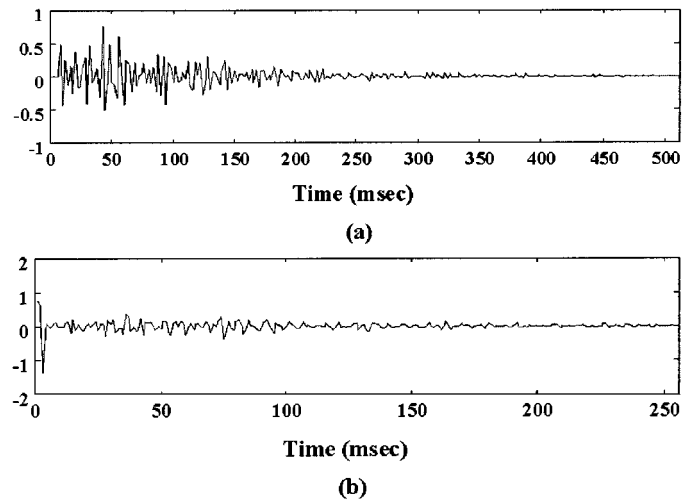


Fig. 6. Impulse responses of (a) the primary path and (b) the secondary path.

the case of $M = 32$, the delayless subband system in [7] requires 882.625 real multiplications per input sample, while the system shown in Fig. 1 is implemented with 771.625 real multiplications. Furthermore, the proposed system in Fig. 3 requires 576.125 real multiplications per input sample. In this example, it was assumed that the wideband filter W is updated every N input samples.

V. SIMULATION RESULTS

To validate the proposed ANC structure, computer simulations were performed. For the simulations, an enclosure with dimensions of $[x, y, z] = [4 \text{ m}, 5 \text{ m}, 3 \text{ m}]$ was assumed. A primary source is positioned at $[0.01x, 0.01y, 0.98z]$. The control source driven by the adaptive filter is located at $[0.98x, 0.98y, 0.98z]$ and the error sensor providing the feedback error signal is located at $[0.90x, 0.90y, 0.90z]$. The primary path and secondary path frequency responses were generated from a modal model of the sound field [1]. The impulse response estimates were then computed from the frequency responses. The primary path was modeled as a 512-tap FIR filter and the secondary path was modeled as a 256-tap FIR filter. The impulse responses of the primary and the secondary path used for the simulation are shown in Fig. 6.

Computer simulations were performed with a broadband signal formed by bandlimiting white noise, using a bandpass filter with a passband from 100 to 350 Hz. Three harmonics (120, 240, and 360 Hz) were added to the reference signal. All signals were sampled at 1000 Hz.

With the generated wideband signal, the performance of the presented algorithm was evaluated and compared with that of the FXLMS algorithm. Figure 7 shows learning curves of the subband ANC systems shown in Figs. 1 and 3. A learning curve of the ANC system proposed by Morgan and Thi [7] is also presented in Fig. 7. The learning curves were ensemble averaged over 100 independent experiments. The convergence parameters were adjusted for each algorithm to have the same variances for the error signal at steady-state. As demonstrated in Fig. 7(b) and (c), both algorithms presented here show faster convergence characteristics than the FXLMS algorithm. Also, it

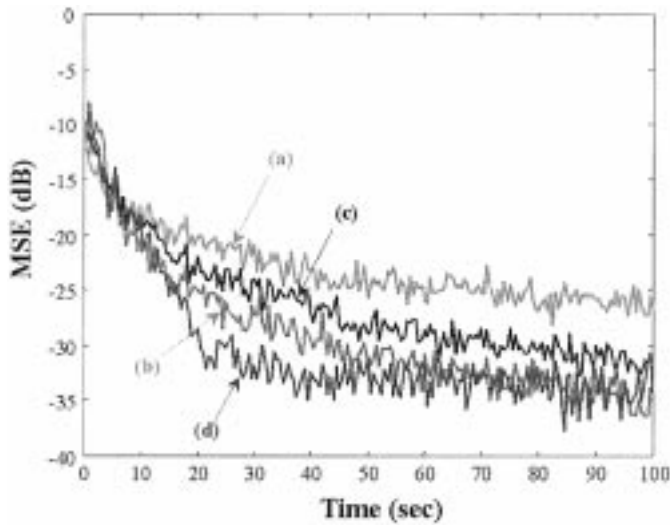


Fig. 7. Learning curves of (a) the fullband FXLMS algorithm, (b) the delayless subband ANC system by Morgan and Thi [7], (c) the ANC system in Fig. 1, and (d) the ANC system in Fig. 3.

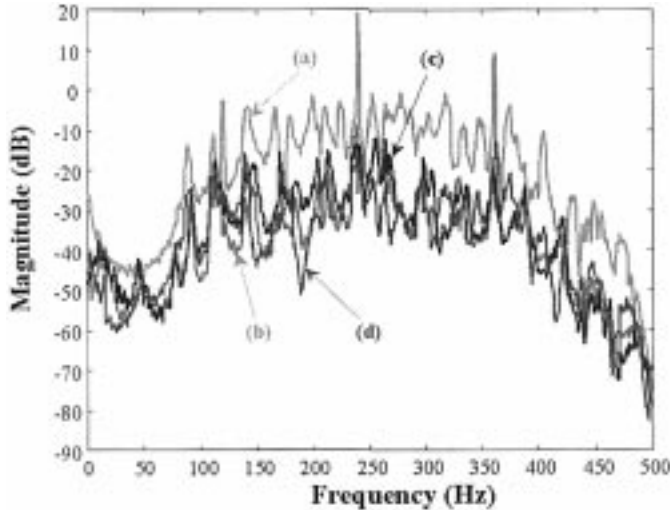


Fig. 8. Steady-state power spectral density of (a) before the ANC system is on, (b) the delayless subband ANC system by Morgan and Thi [7], (c) the ANC system in Fig. 1, and (d) the ANC system in Fig. 3.

is shown that the structure in Fig. 3 has better convergence characteristics than the structures in [7] and Fig. 1. This is because the signal with higher eigenvalue spread ratio can use different convergence parameters which are adaptively normalized with the power of input signal in corresponding frequency band. The steady-state performance of the three algorithms was compared in terms of the power density spectrum of the error signal and is presented in Fig. 8. As shown in (a), (b), and (c), more than 20 dB of noise is cancelled for the whole frequency band in concern.

Fig. 9 shows the impulse response of the secondary path and its estimate obtained using the online system identification approach described in Fig. 2. The result in Fig. 9(b) was acquired by resynthesizing the estimated impulse response for every subband into the wideband. Magnitude and phase responses of both impulse responses are shown in Fig. 9(c) and (d). Although there exist discrepancies between the estimated and actual responses,

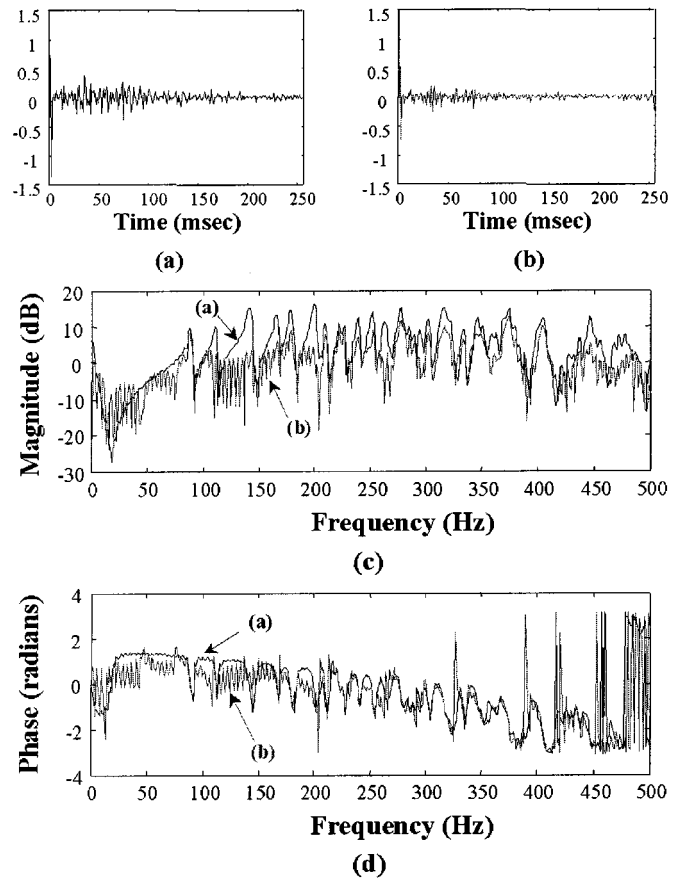


Fig. 9. Simulation results for the delayless subband ANC combined with online identification (Fig. 2). (a) Impulse response of the secondary path. (b) Estimated impulse response. (c) Magnitude and (d) phase responses of the secondary path shown in (a) and (b), respectively.

the overall system was not hindered from converging. With a pure tone reference signal in the limit of slow adaptation, it is known that the estimate of the plant response need only have a phase response within $\pi/2$ of the true phase response of the plant for the system to converge [20]. These results demonstrate that the use of the proposed system provides a reasonable estimate of the true impulse response.

VI. CONCLUSION

In this paper, a new ANC algorithm based on the delayless subband adaptive filter architecture was presented. In the new ANC algorithm, the FXLMS algorithm was implemented in a subband structure with the secondary path transfer function decomposed into a set of subband functions. The secondary path transfer function was then adaptively estimated for each subband in decomposed form. Thus, the structure presented here has significant benefits in computational complexity. Also, the presented structure is apt for the combination with online modeling of the secondary path. In this case, the computational merit could be more significant. In addition, a frequency-domain implementation of the proposed structure further reduces the computational load by employing the block-processing approach. Simulation results were presented to confirm the efficiency of the new ANC algorithm and online system identification scheme presented in this paper.

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